

BIOL3301 Marine Biology

Damian Lau

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Introduction

This course begins with a discussion of marine biology, covering oceanography, shallow marine rocky subtidal systems, coral reef communities, polar communities, pelagic environments, intertidal communities, marine biodiversity, physiology of marine organisms and environmental changes, human impact on the sea and deep-sea communities. This course aims to introduce the whole marine system. Upon completion, students will be familiar with the mechanism and phenomenon in ocean.

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1 Lecture 1 & 2 Oceanography 1 & 2

1. Environment is a major driver of pattern.
 - Physical Environment
 - Biological Environment
2. Oceanography is belongs to physical environment.
3. Temperature: A responds of substrate (Measuring heat)
4. Heat: A kind of energy
5. Water : **Physical Properties**
 - Resist changing temperature when applying heat
 - Very stable temperature
 - It has high heat capacity (Many heat to increase temperature)
 - thermal inertia (Slow Heat up and Cool down)

For the temperature stability, the reason is that water has a high specific heat. It can act as a buffer.
6. the sea temperature has a slow gradual change → stable environment for chemical reaction.
7. Water : **Chemical Properties**
 - Powerful solvent (Universal Solvent)
 - Seawater contains **dissolved solid and gases**
 - Salinity = Total concentration of dissolved **solids**
 - Around 35 ppt salinity = Around **3.5%** dissolved substances
 - Warmer water can contain more dissolved solid = Higher Salinity
 - But! It contains **LESS dissolved gases** ⇒ Higher temperature = Higher Evaporation (Increase the kinetic energy of gas molecules → More likely to escape from water)
∴ Warmer water can dissolve more substances (Solid) but less gases

8. Salinity vs Specific Heat Capacity (Heat-Density) vs Freezing Point

- Ocean: Temperature (-3°C) and Salinity is 35‰
- Estuaries (River): Temperature (-1°C to 1°C) and Salinity is 15‰ to 24.7‰
- Pure water (Fresh): Temperature (4°C) and Salinity is 0‰
- $\uparrow$ Salinity $\Leftrightarrow$ $\downarrow$ Freezing Point $\Leftrightarrow$ $\downarrow$ Heat-Density

9. the physical and chemical properties cause Ocean Stratification (Distinct layer of water)

10. Surface zone is mixed layer (Warmer & Less dense)

11. Pycnocline = Started Stratification

$\Rightarrow$ Rapid temperature change and $\uparrow$ Density

12. A Deeper zone, the cooler and denser environment

13. As I mentioned above, warmer water contains less dissolved gases

$\rightarrow$ CO_2 and O_2 is the most important gases in ocean

- Carbon Dioxide

(a) Ocean = Carbon Sink

(b) $60 \times \text{CO}_2$ in ocean $>$ atmosphere

(c) More dissolve in cold water because warmer water is less dissolve gases

(d) Speed of CO_2 moving towards from atmosphere to ocean $>$ Speed escapes from ocean to atmosphere
 $\Rightarrow$ Ocean absorbs CO_2

(e) After entering water, CO_2 converts to Hydrogen Ions (H^+) and Carbonate Ions (CO_3^{-2}) (Equilibrium)

(f) $\uparrow \text{CO}_2 \Leftrightarrow \downarrow \text{pH Value}$ and cause Ocean Acidification (i.e. $\text{pH} = 7.5 - 8.5$ and keep reducing by ocean acidification)

$\therefore \text{H}^+$ Ions lowers the ocean's pH

- (g) Autotrophs : Get carbon from CO₂ (Inorganic Chemical)
- (h) Autotrophs = Capture light energy and convert light into nutritional organic substances (Can produce its own food)
⇒ E.g. **Phytoplankton** and **Algae**
- Oxygen
 - Essential for life
 - Diffusion rate (Speed) in water is 10000 greater in air
 - Carrying capacity is < 1% in water vs 21% in air
 - Oxygen produced in the ocean = Oxygen produced in terrestrial
⇒ But Ocean has 300X less photosynthetic biomass
 - Solution : Phytoplankton & Autotrophs helps the oxygen and organic nutrients productivity
- Adaptation for Species (To get more O₂)
 - Pumping water across respiratory surfaces (Sharks : Keep swimming for pumping water)
 - Live in high water velocity (More water flow, More O₂ can pass through)
 - Specific gas exchange structures

14. CO₂ vs O₂ vs Depth

- (a) O₂ is **relative high** in the surface zone is because it has sunlight. This means that the **biomass is highest** and plants can do photosynthesis in this zone
- (b) ↑ Depth → ↓ O₂
∴ No sunlight and all species are started to do respiration rather than photosynthesis
- (c) there is slightly increase after 1000m below the surface
 - Reason 1 : Temperature is very low ⇔ Lose of biomass that using O₂
 - Reason 2 : Temperature is very low ⇔ O₂ is more dissolvable

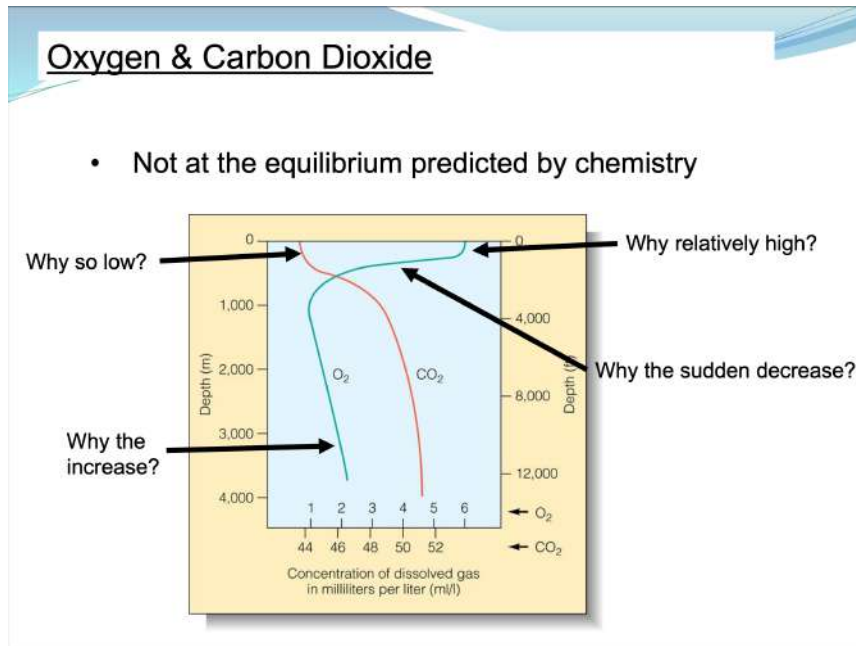


Figure 1: Gases Level Change

Primary Productivity = Autotrophs produce their own organic compounds (food) from inorganic substances by converting light energy

⇒ Use Photosynthesis to get organic matter (**Main Product**)

⇒ At the same time, Using sunlight to get **byproduct O₂** in Photosynthesis

1. Primary Productivity Byproduct + Limiting Factors in Ocean

- Oxygen (O₂)
 - When the plant is at the water surface, it has higher photosynthesis rate and highest **Net Productivity** (Closer to the sun)
 - But photosynthesis will be inhibited because light is too intensive that UV light will damage the plant
 - **Net** Productivity arrive 0 at compensation depth
 - Compensation Depth : Respiration = Photosynthesis → Just ok to do metabolism
 - ⇒ This means **No Growth + No Net Lost**
 - Below the **Compensation Depth**, as light levels decrease, Respiration dominates and Leading the environment to a lower O₂ level
- Light Extinction
 - the shallow layer absorbing the red light quickest and so on
 - Restrict some species distribution
 - E.g. Green plants cannot live in deep water; But benthic algae and corals can live in 50 m

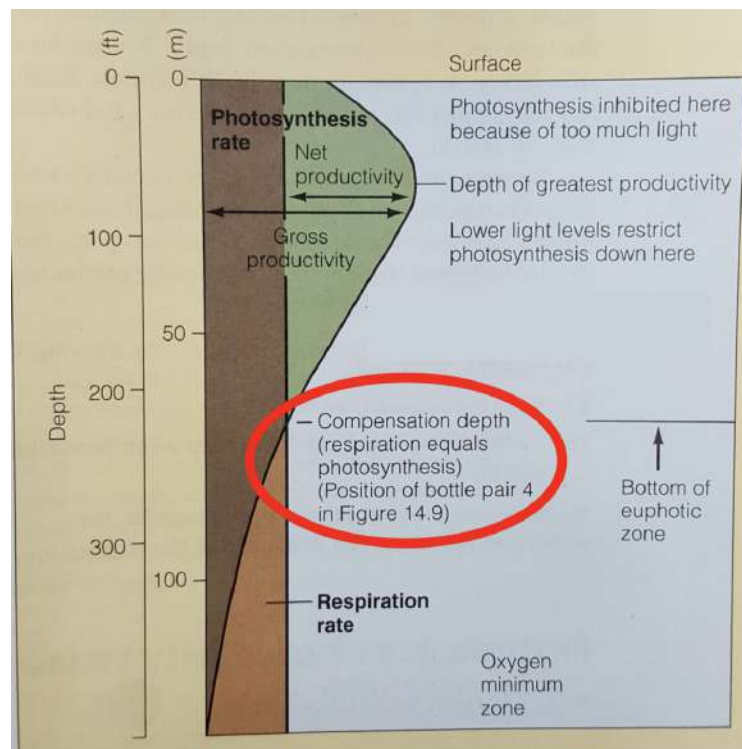


Figure 2: Light & O₂ vs Depth

- Nutrients
 - Hard to measure nutrient concentrations in ocean
 - Taken up by photosynthetic organisms rapidly
 - Use **chlorophyll- α concentration** for the replaced representative measurement of nutrients
 - the most productive zone is **Upwelling Areas** → Brings lots of nutrients → Rapid Phytoplankton (Autotroph) Blooms
- 2. Ocean Circulation: Wind and Weather (Typhoon) are driven by Sea Surface Temperature
 ⇒ Evaporation → ∴ Rotation of Earth → Temperature Change (Heat Flow)
- 3. Ocean Current : Primarily driven by **Wind (Surface Current) and thermohaline Gradients** (Meaning Seawater Density Change is caused by Temperature + Salinity + Depth (Pressure)) (**Deep current**)

4. **Coriolis Effect** : Creating **opposite rotation directions of wind (Typhoon) and Ocean Currents**
⇒ Form Ekman Spiral
5. Closer to equator, Lesser Coriolis Effect & Ekman spiral effects
 - ∴ Earth Self Rotation
 - Northern Hemisphere : **Deflect** an angle to the right perpendicular direction
 - Southern Hemisphere: **Deflect** an angle to the left perpendicular direction
 - Surface Water : Smaller Deflect Angle
 - Deeper Water : Larger Deflect Angle (Almost 180° = Not **Net Movement**)
6. **Ekman Transport** : **Net Movement** of ocean current to approximately 90° affected by **Wind + Coriolis Effects**
 - Northern Hemisphere : 90° turn right of the wind direction
 - Southern Hemisphere : 90° turn left of the wind direction
 - Ekman spiral : the water current bent due to the depth of water (Deeper, Bent more)
 - Depth ↑ ⇔ Water movement ↓ ⇔ ↑ Deflect angle
7. Thermohaline Circulation (the Great Ocean Conveyor Belt)
 - Warmer Surface Flow : Near the Terrestrial
 - Cool Subsurface Flow : Near the Polar
 - Deep Current : Driven by differences in **Temperature & Salinity of water masses** = Cooler + Higher Salinity
 - Shallow Current : Warmer + Lower Salinity
8. **Upwelling**
 - Transfer deep, cold, nutrient-rich water to the surface layers
⇒ Rapid Phytoplankton Blooms
 - Driven by wind blowing surface water offshore

- Small Scaled, Localized, Not frequently happened
- Water Move = **Friction** between wind and water surface
∴ Water Movement = Surface Friction
- Current Circulation : Upwelling + Coriolis Effect (Controlling Upwelling Direction) + Ekman Transport

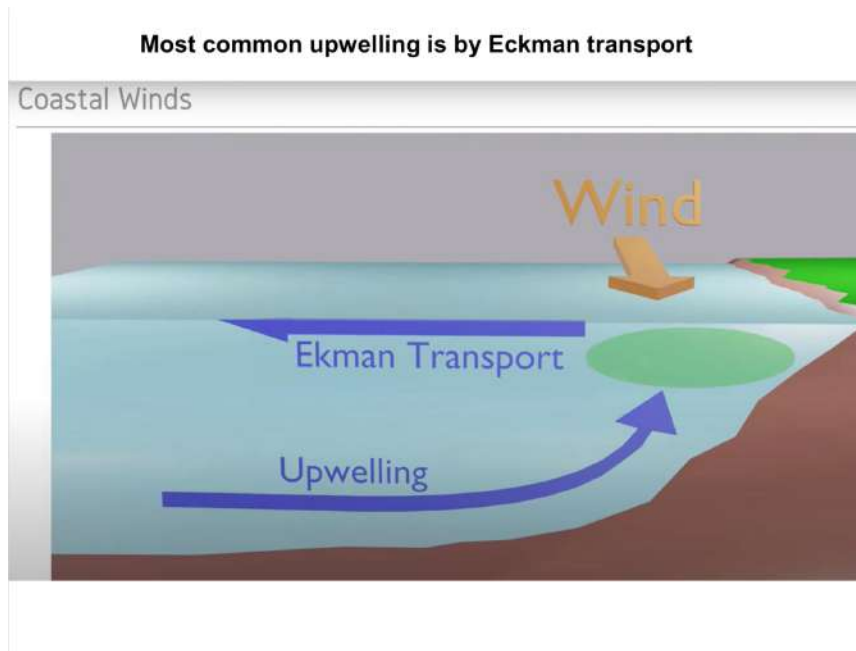


Figure 3: Upwelling

9. Categorize Marine Environment

- Natural Descriptors
 - Abiotic
 - Geological
 - Biological
 - Geographical (Location)
- Anthropogenic Descriptors (Human-Being)

10. Geological Zonation

- Neritic Zone : Most Biology happen here (Upper Photic Zone)

- Continental Shelf (Kind of Neritic Zone) : Only 7% – 8% of ocean area

11. Structure of Marine Environment

- Coastal
 - Estuarine (Sediments)
 - Seagrass
 - Mudflats
 - Reef (Rocky and Coral)
 - Rocky Shores (Intertidal)
 - Sandy Beaches
 - ⇒ Both Terrestrial & Marine Characteristics : Rocky Shore + Sandy Beaches
- Open Ocean
 - Continental Margins (Terrestrial Margin)
 - Pelagic Zone (Almost Deep Sea)
- Deep-Sea Habitats

12. Biotic Habitats (Commensalism)

- Biotic Factors
 - (a) Have dominant organisms
 - (b) Habitat forms the basis of ecosystems
- Example
 - (a) Rocky Reefs vs Algal Canopies
 - (b) tropical Reefs vs Hard Corals
 - (c) Sandy Sediment vs Seagrass

13. Large Marine Ecosystems (LMEs)

- Areas: $> 20,000 \text{ km}^2$
- Ecology that driven by currents, upwelling and connectivity
- Most highly productive ocean are supported by primary productivity of coastal ecosystems globally

2 Lecture 3 & 4 Oceanography 3 & Subtidal Rocky Habitat

1. Connectivity

- Interaction between Populations and Habitats
- Large Capacity for migration between population
- Depending on larval longevity (Life age), larvae behaviour, water movement and availability of habitat
- Mainly Open system (Population is not self-sustaining → Interdependence)
- Can be affected by human activity and natural processes

2. Population Connectivity

- Dispersal is important for population increase
- If live closer to their own parent,
Advantage
 - Increase SurvivalDisadvantage
 - Have competition with their Parents
 - Predators can easy to find
 - ↓ Genetic Evolution
 - ∴ Not need to adapt a new environment by themselves

Limiting Factors

- Currents
- Biological Traits
- Habitat Availability

3. When we consider a specific place's population connectivity, we should focus and consider different phase of species
i.e. Some species have different habitat and behaviour in larval phase and adult phase

4. Enhance Connectivity

- Coastal Hardening (Species can attach on it and settle down)

- Shipping (More in Business ship)
 - Normally have a biogeographic break (Natural Barrier)
 - Human Activities break the isolation and species can move across the break
 - E.g. Stick on the ship
- Boating (More in Individual and **Regional**)

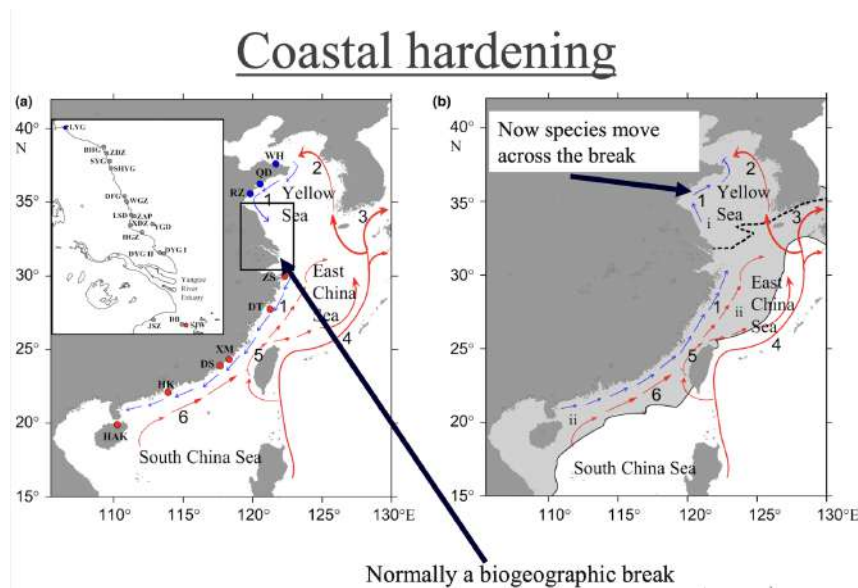


Figure 4: Breaking Natural Barrier

5. Classic Invader Condition

- Demography
 - (a) Large Biogeographic
 - (b) Competitively Dominant
- Reproduction
 - (a) Asexual + Sexual
 - ⇒ More complex reproduction method
 - (b) High Reproduction Rate + Good Dispersal Ability
 - ⇒ Hard to endanger and get rid of
 - (c) Short Generation Time (Cost of death is low)
 - ⇒ You already have your next generation before you die

(d) Fast Growth Rate

⇒ Become sexual mature as fast as possible

- Physical

(a) Good ability to overcome salinity condition

- Tolerance

(a) Good tolerant of temperature

(b) Good tolerance of disturbance

- Behaviour

(a) Associate with Humans

⇒ Human is the most success and dominant species in Earth

(b) Gregarious (Live in family unit)

⇒ Higher chance to get success in reproduction

- Diet

(a) Board

- Still cannot predict the species places or timing of introduction

6. Vectors of Introduction

- **Transports large volumes of water rapidly** from coastal habitat

⇒ E.g. Artificial Shorelines

- **Ballast water from cargo ships** (Have a chance to get species to another place)

- Solution : Legislation = Pump local water into cargo

⇒ Ensure the cargo don't have any foreign species entering

- Hull fouling

⇒ Ship is broken

- Intentional release : Mercy release

⇒ Like buy a fish from wet market and release them to the ocean / river

7. Shipping Impact

- High probability of invasive species being separated during cargo transportation
- Mainly in Asia country and China

8. Ballast

- 95% are from cargo carriers
- Mostly the marine invertebrates and mainly in larval phase
∴ Larvae = Free-floating in the water column = Can enter ballast water easily

9. Effects of Introducing Predator

- (a) It makes a fake natural system
- (b) Productive fisheries bring 2 new fishes to the new habitat
- (c) Zooplankton nearly totally utilized by Planktivorous fish
⇒ Break the balance of that habitat +
Fishes become the invasive species because of the economic activity by fishermen
- (d) Special Sudden Case : Freshwater-flow is reduced + Increased pollution
⇒ Leading to destruction
- (e) It increases the salinity (Less freshwater) and a jellyfish (*Scyphomedusa*)
⇒ ↓ Zooplankton (Be eaten and cannot survive in high salinity conditions)
- (f) Indirectly, Fish Production also reduced
⇒ Since fishes eat Zooplankton
- (g) Government increase freshwater to eliminate the invasive species *Scyphomedusa*
- (h) But another new invasive jellyfish bloom out and kill 82% of Zooplankton
- (i) Eventually, Devastated 2 fisheries (Even more worse)

This means eliminating an invasive species cannot use another species to control and monitor, or don't even let foreign species enter other habitat easily

10. Intentional Introduction (Naive Feel to Environment)

- It would destroy the native habitat
- May facilitate other new invasive species
- Influence is unpredictable
⇒ E.g. Mercy release might have unknown effect

11. We have good ability to let things spread and come in but have poor ability to predict the species places and times of species introduction (the continuous influence)

Rocky Habitats

1. Subtidal : Below tides and always under water
2. Also, refer to Neritic Zone (Nearshore)
3. Have Sunlight (Can do photosynthesis and drives system productivity)
4. Photic Zone (Shallow Subtidal) : Greatest Sunlight $\Leftrightarrow$ Greatest abundance of photosynthetic organisms
5. Recall : Compensation Depth $\rightarrow$ Oxygen Production = Oxygen Consumption = No Net Change
6. Light Penetration sets the lower depth limits of photosynthetic species
 $\Rightarrow$ Photosynthetic Species can only grow above the compensation depth
7. Algal Forests
 - Macroalgal Forests
 - Brown Algae (Phaeophyta)
 - Attached to rock (Benthic) $\rightarrow$ Kelp Holdfast
 - Simple Reproduction
 - 2-phase life cycle
 - Extensive subtidal and intertidal species habitat
 - the Shapes and Size based on the dominant species
 - Seasonal Forests in tropical region (But grow rapidly)
 - General Diverse (Mosaic) Structure : Forest + Turfs + Urchin Barrens
 - Growth Period : Around November (Generally)
 - Environment : $\downarrow$ Temperature $\Leftrightarrow$ $\uparrow$ Number of algae
 $\Rightarrow$ Global Warming shorten their life time
 - they are productive
 - Similar function to rocky subtidal ecosystem

- Temperate Coastal Waters : Around 25% of world's coast-lines = Great Biomass cycle

8. Resources for Humans

- Food Resources
 - Algal Harvesting (Direct)
 - Fisheries (Indirect)
- Human Activity may broke the above forest cycle and cause ecosystem shift

9. Relationship between Canopy Algae vs Turf Algae vs Urchin Barren (Coralline Barren)

- (a) In South and West Australia, Canopy-forming algae (Kelp) is dominant species and **Non of Urchin Barren** (Kind of Coralline Barrens)
- (b) Turf Algae is not the dominant one
- (c) Reason : West is more cold and it is good for the Canopy Kelp Algae to grow
- (d) In East Australia, Turf Algae and Urchin Barrier grow exponentially $\Leftrightarrow$ Canopy Algae decreases
- (e) Reason 1 : Brown algae (Kelp) is **benthic** and loves to attach the rock, where urchin barrens will clean the rock and become shiny rock
- (f) Reason 2 : the shiny rock allows Turf Algae growth
- (g) Reason 3 : Turf Algae also inhibit the growth of Canopy Kelp Algae
- (h) All species don't like to go out their habitat

10. The above is a **Top-Down Driven System**

$\Rightarrow$ Meaning the abundance of organisms are **controlled by a higher tropical level organisms** = The Sea Urchin

11. Solution Steps to Top-Down Driven System

- (a) Reserve (Protected Area by Human)

- (b) Increase Carnivores
 - (c) Decline sea urchin and invertebrate grazers
 - (d) Rebuilt of Canopy Kelp Forest and replace the empty barrens
12. Also a method to protect some endangered species (But it might have some side effects)
13. Bottom-up Driven System
- (a) Higher Trophic Levels are regulated by **the abundance of resources and quality** at Lower Trophic Levels
⇒ Not only species, "Low Trophic Level" can also be physical factors : Lights, Oxygen
 - (b) Limiting factors : Light & Nutrient
 - (c) Importance : Resource Availability > Predator-Prey Interactions
14. Kelp Loss in Southern Australia
- Increase Population (Excessive Nutrient Loads) and Coastal Housing
 - Faster decline with more rapid population growth
 - Alternate state maintained by altered conditions
 - May link to influence of urban catchments (Likely affecting water quality and ecosystem health)
 - Higher human population density leads to the increase **nitrogen levels in water**, which in turn contribute to significant Canopy Kelp Algae loss in reef ecosystems
 - Excessive nutrient loads
 - (a) Increase nutrient and sediment favour growth of Turf Algae
 - (b) Turf Algae inhibit recruitment of Algae Kelp Forests (More Complex form) → Negative impact
 - (c) This is the **Bottom-Up Driven System**

15. Reason why Canopy Kelp Forest cannot replace by Algal Turf

- (a) Kelp have a bi-phasic (2 phase) life cycle
⇒ Can spread into male gametophyte and female gametophyte
- (b) Exceed nutrient will kill Kelp's gametophyte and young sporophyte
- (c) Increased nutrients and sediment also favor algal Turfs, which has better adaptation than Kelp (as mentioned in 14(b))
- (d) then it is disrupting Canopy Kelp life cycle by **inhibiting gametophyte development**
- (e) Turf Algae become dominant species
- (f) Eventually, leading to Canopy Kelp Loss.

16. Solution Steps to Bottom-Up Driven System

- (a) Stop pumping pollution (Decrease the **Exceed** nutrient)
⇒ Allow more female gametophytes and young sporophytes survive
- (b) Inhibit the growth of Turf Algae
- (c) Leading to the rebuilt of Canopy Kelp Forest

3 Lecture 5 & 6 Coral reefs & Polar Communities

1. Coral Type : Cnidaria (Phylum)

- **Scleractinians** : Hard Coral = Reef = Building Corals
- **Actinarians** : Anemones (Soft Tissue)
⇒ Not a Coral
- **Alcyonaceans** : Soft Corals

2. Coral : Holobiont

- Have symbiosis with other organism
- ⇒ Consisting of a **Host Organism**

3. Found in Tropical and Subtropical Environments

- ⇒ Hot Spot : Indo-Pacific & Coral Triangle
- ⇒ Most abundant : East Countries

4. Physical factor limiting distribution

- Temperature ($20 - 28^{\circ}\text{C}$)
 - (a) Warm water = Rapid calcification
 - (b) Too hot / Too cold = Die
- Depth & Light
 - (a) The symbiosis depending on light
 - (b) Light Intensity : Too weak
- Salinity : Coral grows well in high salinity
- Turbidity & Sedimentation
 - (a) Dinoflagellates require light
 - (b) Need clear water (Usually at low nutrient tropical seas area)
 - (c) Sediment inhibit feeding
 - (d) Too much food → **Smother Polyps**
⇒ Polyps cannot breath
- Wave Energy
 - (a) Damage and fragment corals

5. Symbiosis : Tropical Corals v.s. Symbiodiniaceae
 - (a) Obligate Symbiosis
 - (b) Symbiosis can let corals adapt well in Oligotrophic Tropical Water
⇒ Oligotrophic = Less Nutrients
 - (c) Symbiodiniaceae is used for 100% metabolic requirement in corals
6. Get nutrients on plankton heterotrophically
E.g. Zooplankton (Low Abundance)
7. Heterotroph = An organism that cannot produce its own food
8. Corals use stinging cell (Cindocysts) in tentacles to eat
9. Symbiotic Dinoflagellates (Symbiosis with Coral)
 - Unicellular Algae
 - Living in endotherm cnidarians
 - Abundant in corals
 - Nutrients and Adaptation that can get from symbiosis
 - Carbon, Nitrogen, Phosphate, Carbon Dioxide
 - Need Light
 - Habitat
10. Nutrients and Adaptation that can get from symbiosis
 - Glucose, Amino Acid, Lipid
 - Oxygen
 - ↑ Rate of Calcium Carbonate Deposition (↑ pH = Calcification site)
11. Reef-Forming/ Building Coral = Group of coral held together by calcium carbonate
 - (a) Massive
 - Resistant to storm damage

- Slow recover rate
- (b) Branching
- Easy to damage by storm
 - Fast Recover Rate

12. Reproduction

- Broadcast Spawner : Release packets of sperm + Eggs into water
- Maintain **synchronous spawning** (Reproduce together at the same time)
⇒ ↑ Chance of Fertilization
- **Asynchronous Spawning (Different time fertilization)**
⇒ **Reproductive Isolation** + **Speciation** = New Species

13. Overall Steps of Coral Reproduction

- (a) Developed Larvae sink to benthos (Bottom) and Metamorphose (Change the Morphology)
- (b) High Death Rate (Mortality) in recruits (New corals) : Find good settle place by strong selective pressure
- (c) Single **polyp** juveniles (Coral Larvae) : Produce large clonal coral colony by **asexually**

14. Detailed Steps of Coral Reproduction

- (a) Release packets of sperm + Fertilized Eggs into water
- (b) Cell cleave and zygote develop to larvae
- (c) Larvae drift with current and find suitable substrate to settle
- (d) Larvae metamorphosis can form mouth and tentacles from polyp form
- (e) Recruit lay down skeleton and take up **algal symbiosis** from environment
- (f) Polyp budding grow to juvenile colonies, and then grow large enough asexually to become sexually mature

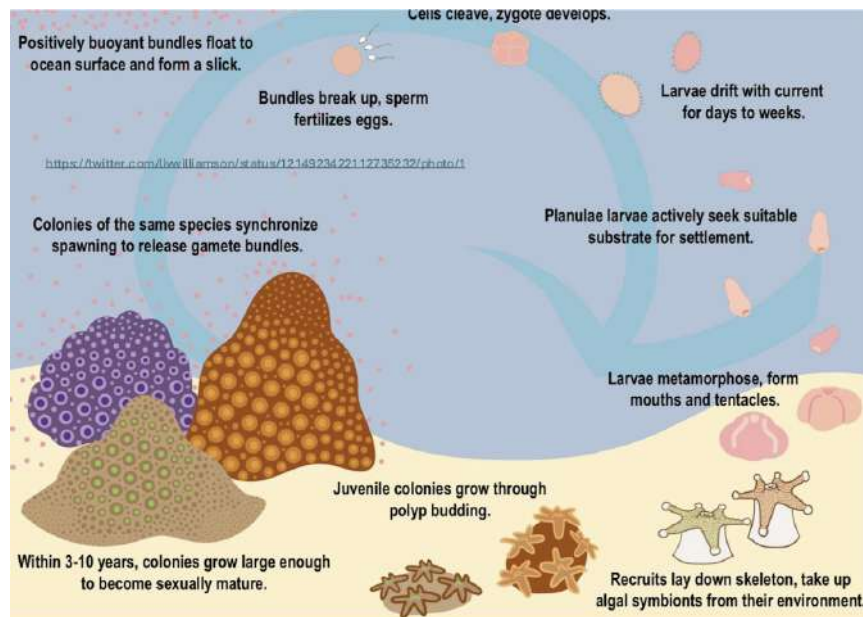


Figure 5: Coral Reproduction

15. Build a Coral Reef

- (a) Framework Starting Material : Coral, Coralline Algae, Serpulid polychaete and other CaCO_3 secreting organisms
- (b) Erosion and Fragmentation : Some framework is **Broken by storm and Eroded** by clams, worms and boring sponges
- (c) Mortar : Coral fragments and sediments are bound together by **Calcareous Algae** (Stick Together)
- (d) Lithification (Become Rocky) : **Magnesium Calcite** is Precipitated (Sink) and Cementing (Stick) the reef together
 $\Rightarrow$ This process transforms dead coral structures into fossilized rock
- (e) Eventually, Fragments consolidated into reef structure
- (f) Reef Growth Importance : Frame Building > Erosion
 $\therefore$ Frame Building determines the reef's ability to grow and adapt
- (g) If Erosion > Frame Building, then it would leads to degradation

16. Coral Reefs Zonation (Separation Zone)

- Very diverse in different water level
- Shallow = Get more light ; Deep = Avoid storm
- Support 25% marine species

17. Threats to Coral Reef

(a) Predation & Bioerosion of Corals

- Population explosions cause coral decline
- Sea Star eat corals
- then sea urchins can attach to the rocks (Last Lecture)

(b) Human Impacts

- Over-fishing of fish
- Urchin population explosion
- Diseases & Mass of Urchin Mortality
- Algal domination & Coral loss (Have Algae $\Leftrightarrow$ No Corals)
∴ Algae can attach to the rocks

(c) Increase Water Temperature

(d) Extreme thermal Events (Global Warming + Heatwaves)

(e) Ocean Acidification (CaCO_3 is Alkaline and Calcification requires high pH environment)

(f) Sedimentation (Inhibit Feedings) and Turbidity (Fragment Corals)

18. Coral Bleaching (Become White)

- thermal Stress : Corals and their Symbionts are disassociated
⇒ Cannot too hot
- thermal stress → Symbionts produce an excess **Toxic** Free Radical → Expelled out of the Coral Host Tissues
- Corals can survive without their symbionts for a short period
⇒ Ultimately die if no regain their symbiosis

- When coral dies : Filamentous algae grows immediately on the skeleton

19. How to Manage ?

- Set Marine Protected Area (MPA) : But cannot against global changes (Global Warming)
- Coral Intervention
 - (a) Asexual propagation
 - Advantage : Relatively easy
 - Disadvantage : Adaptive capacity is limited
 - (b) Assisted Gene Flow
 - Breed with warm adapted corals sexually with warm sensitive corals
 - Spread the adaptive alleles

20. Limits of Coral Restoration

- Expensive and Extremely Labor Intensive
- Only able to target specific species
- Bring individuals away from home : Removing survivors to the place that they are adapted
 - ⇒ Natural Selection : Already choose the strongest individuals at that habitat

Polar Ecosystems

1. General Traits of Polar Regions

- Cold
- Polar Light : 24-hour Daylight (No Night) throughout the Summer
⇒ Polar Night : 24-hour Darkness (No Light) throughout the Winter
- Extreme changes in ice cover
- Photosynthetic Base of Food Web : Diverse Phytoplankton and Algae Community
- Area : Arctic (North) and Antarctic (South)
⇒ They have different environment

2. Adaptations at Animal Level

- Thick layer of Fat
⇒ For Isolation
- Oily skin & Fur
⇒ Repel cold water
- Small ratio of Surface Area to Volume ⇒ Minimize Heat Loss
- Slow Growth Rates
- Low Basal Metabolism
- ↓ Annual Reproductive Effort relatively

3. Adaptations at Molecular Level (Bacteria)

- Structures : Cold-shock Domain
- Cold-shock Proteins
 - Compensated : For cell-membrane fluidity
 - Maintain activation of enzyme at low temperature
- Cryoprotectants
 - Prevent cells and tissues damage when freezing
E.g. Anti-freeze Glycoproteins (AFGP)

- **Anti-freeze Glycoproteins (AFGP)**: Maintaining fluidity of blood
- Bind to ice crystals

4. Arctic Sea

- Latitude of Arctic Circle : 66° North
- Winter : Ice cover the ocean
- Summer : Ice melt majority and contain Higher Primary Productivity from **Phytoplankton**
 - ⇒ Mainly **Large Diatoms** → Process abundant organic matter
 - ⇒ Strong Benthic-Pelagic Coupling
- Melt ice can release of sand particles (Source of **Metal and Micro-nutrient**)
- High Primary Productivity
 - ⇒ **Copepods** : Eaten by planktivorous fish
- Algae bounded by ice
- Provide organic matter to benthos

5. Ice algae

- Mainly grow in spring / fall exponentially (Low light but high nutrient)
- Grow under the ice and the porosity
- Porosity (Holes) : Can trap nutrients + light

6. Adaptations of Ice Algae

- (a) Ice binding proteins : Form a gelatinous (Gel like) cover to protect from ice
- (b) Xanthophyll : Prevent photodamage
- (c) Uneaten Algae and Phytoplankton will go to the benthos (Sea Floor) : Feed large benthic invertebrates
 - ⇒ Bivalves, polychaetes, fishes

(d) **Bentho-Pelagic Coupling** : Organic matter and nutrients go from water column to the benthos

⇒ Resuspend particles + Water → Back to surface

(e) New Formation of Sea Ice : Would change the depth of the **pycnocline**

⇒ then cause **additional mixing**

7. Climate change in Arctic

- Less ice = Higher water temperatures
⇒ Light can easily penetrate → More warming
⇒ Less sea-ice algae aggregates reach the bottom
- Affects organisms that depend on ice for their daily activities
⇒ Polar bears use ice for hunting
⇒ Seals use ice for resting
⇒ Ivory Gull use ice for breeding ⇒ Species need to move upwards, invading Arctic regions

8. Invasions into the Arctic Ocean (Fishes and Invertebrates)

- (a) Polar cod (Arctic Pod) : Eat zooplankton and eaten by birds, seals, whales
- (b) Atlantic cod : **Invade** polar waters due to global warming (Might be competition for resources)
⇒ Can tolerate warmer temperatures better than Polar Cod

9. Climate change in the Arctic (Salinity)

10. Great Salinity Anomalies (GSA)

- (a) Lower Salinity : More ice melt in spring and summer
∴ Diluted
- (b) Also affecting stratification (Divided Layer)
- (c) Slowing down of the Global Conveyor belt

11. Arctic vs Antarctic

- (a) Arctic

- More harsh conditions
- Low richness
- Infauna (Within)

(b) Antarctic

- Stable conditions
- Deep surrounding ocean = Isolation
- High richness
- Epifauna (Upon)

12. Antarctic Ecosystems

- Large seabird population: phosphorus and nitrogen in summer can be obtained from abundant guano (Bird PoPo)
- Ice extends from land to the sea
- Land remains covered in ice in summer
⇒ Sea Ice will melt
- Melting ice can release nutrient in sea ice

13. Food Webs

- (a) Krill (Keystone Species) : Dominate zooplankton
- (b) Feed on ice algae
- (c) Ice can provide protection
- (d) Abundance and Distribution = Ice area
- (e) Penguins and whales feed on krill
- (f) Krill accumulate nutrients and metals and then eaten by large species
- (g) Nutrients are released back to the water column when organism **defecate**

14. Human Activity (Commercially / Pollution) also affect the food web

15. Krill are used for dietary supplement for many diseases
⇒ E.g. Heart Disease Medicine, High Blood Pressure, Aging Skin

16. Antarctic Circumpolar Current (ACC)

- Strongest water current in the world
- Maintain water temperature
- Create connectivity between distant area (South America and New Zealand)
- Acts like a Barrier between populations in Antarctica and adjacent
⇒ Lead to speciation and **endemism** in Arctic
- Some species are able to cross the ACC but still an efficient barrier
- Endemism: Species that is only found in a specific geographic location

17. Antarctic Diversity

(a) Shallow water area

- Physically isolated with other ocean area
- Because of **ACC & Geographic position**
- Very high diversity and endemism
- Dominant animals : **Suspension feeders** (Sponges, Bryozonas and Hydrozoans)

(b) Deep water area + Benthic Area

- Higher diversity of benthic organisms than Arctic
- Dominant animals : Mobile predators (Brittle Sars, Sea Stars, Isopods and Gastropods)
- 50% of them are invertebrates (Benthic)
- Size is much larger than other place

(c) Both lack of shell-crushing predators (Lobsters, Crabs)

∴ No predators can break the shell of gastropods

(d) No dynamic weather (Due to the Antarctic Circumpolar Current)

(e) Area closed to Antarctic : Influence by South America and New Zealand

18. Evolutionary in Antarctic Diversity

- (a) 88% fish species only in Antarctic
- (b) Cod Ice fishes (Nototheniidae) are very diverse in [Southern Ocean](#)
- (c) Recall : Antifreeze Glycoprotein (AFGP)
 - Maintaining fluidity of blood
 - Bind to ice crystals
 - Keep fish to be small
 - Avoid the total freezing
 - Prevent ice crystals from melting (So they can move everywhere all the year)

19. Climate change and Human Influence in Antarctic

- (a) Animal Issue
 - i. Krill
 - Ice algae has declined with the melting of ice, affecting krill recruitment (No food)
 - Southern Ocean Krill is decreasing (Population are moving to polar area (Can only live in under ice)
 - Krill overfishing (Have High concentration of Omega-3 oils)
 - ii. Adelide Penguins
 - Decrease in population density
 - Need ice for nesting
 - Krill Decrease $> 50\%$ (Less food) $\rightarrow$ Diet is not flexible
 - iii. Gentoo Penguins
 - Don't affect a lot
 - Diet is flexible
 - Don't require ice for nests
 - Can dive deeper (Find more food/ Against predators)
 - iv. Invasions during Global Warming

- Warm temperature affect metabolism and calcification of Antarctic animals
- Increase the density of **King Crab** : A shell break predator
- King Crab : Cannot live in cold water (Disturb the Mg⁺ ions balance)
- ↑ Temperature, Crabs can dominate Benthic ecosystems (Antarctic species cannot defend as mentioned above)

(b) Antarctic Ocean Acidification

- i. Antarctic Pterapod (Limacina) : **A Planktonic snail** and very abundance
- ii. Dead → Sink Fast → **Export carbon to deep ocean** significantly
- iii. **Aragonite shell** : Very sensitive to ΔpH in surface waters
⇒ Increase Death Rate (Dissolution) + Malformation (Strange Shape Formation) of Shells

20. Threats to Both Polar Regions

(a) Human Activity Factor

- Oil drilling close to poles : ↑ CO₂ emissions + Risk of oil leakage
- Overfishing (Krill, Fish)
- Increase transition around polar region : Increase pollution and invasion (Ballast water)
⇒ Mentioned above

(b) Global Warming Factor

- ↑ Temperature → Species Invasions
- Loss of Ice Coverage : Affects Primary Productivity and Benthic Food Web
- Increase Metabolic Demand : Species adapt the cold-water condition (No ice to rest)
- ↑ Transition around Polar Region

4 Lecture 7 & 8 Pelagic Environments & Benthic Communities

1. Plankton

- Organism live in water column
- Too small to swim against the ocean
- Passive Organisms → But still can swim

2. Classification 1 : What they eat

- **Phytoplankton** : Photosynthetic bacteria (Primary Producer)
- **Zooplankton** : Non-photosynthetic organism + Eat Plankton
- **Mixotrophs** : Can do photosynthesis and eat plankton

3. Classification 2 : Life stage of Plankton

- **Meroplankton** : Only part of the time (Juvenile Stage) is **Zooplankton**
⇒ E.g. Fish will become a strong swimmer
⇒ E.g. Frog can leave water
- **Holoplankton** : Whole life in **water column and in plankton form**

4. Classification 3 : Water & Air Interface

- **Neuston** : Floating on the water & air interface
⇒ E.g. Algae & Bacterial mats at surface
- **Pleuston** : Organism that live in water can float out the water (**Protrude**)
⇒ E.g. Manowar

5. Classification 4 : Based on size

⇒ Size : Femto < Pico < Nano < Micro < Meso < Macro (-plankton)

6. Main Categories of Plankton

(a) Marine Viruses

- Size : Similar to Femtoplankton (Smallest Plankton)

- Made by DNA or RNA strands (Protected by **Protein Coating**)
- Very small
- No metabolic activity
- Require a host to survive
- **Replicate rapidly** by bursting the host cells
- Can contribute significant to food web and nutrient cycling
- Control the **Density of Bacterioplankton and Phytoplankton**
⇒ Can promote Coexistence

(b) Bacteria & Archaea

- Picoplankton (2nd small)
- Most marine bacteria : **Gram Negative** = Have Cell Membrane → Protect from harsh conditions
- **Reproduce rapidly** when there is high abundance of organic matter
- Essential for nutrient cycling (Can take carbon quickly)
⇒ Then eaten by higher level of consumers and predators
⇒ Nutrient can return to higher trophic level
- Aerobic Decomposition (When there is oxygen)
- Example of Anaerobic Bacteria (Anoxic condition)
 - i. Methanogenic Archaea : Use CO₂ for respiration to produce Methane CH₄
 - ii. Denitrifying Bacteria : Reduce nitrate to N₂ (Gas State)
- **Phytoplankton** & Bacteria vs Viruses
 - i. Break the cell
 - ii. Release organic matter to water
- Microbial Loop

The Microbial Loop

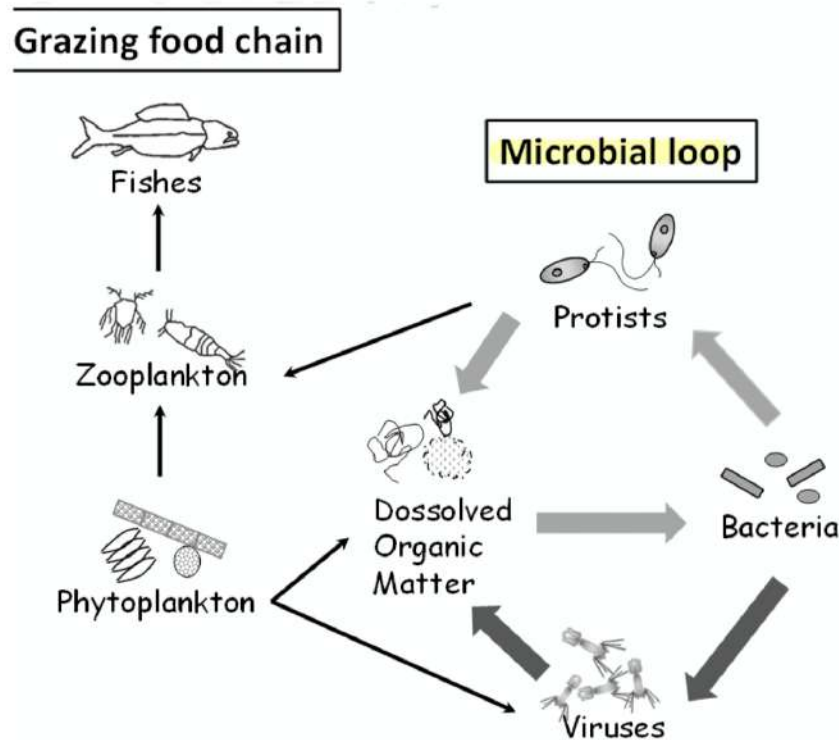


Figure 6: Microbial Loop

- Can affected by Viruses
 - i. Break the cell
 - ii. Releases organic matter to the water column
⇒ **Fueling & Providing energy** to the cycle
- Examples of Bacteria
 - i. Cyanobacteria = Photosynthetic Bacteria
 - Common Name = Blue-green Algae
 - Most dominant organism in ocean
 - Function 1 : Produce 50% of O₂ (By-product) in Earth
 - Function 2 : Nitrogen Fixation (Convert N₂ to NH₄)
⇒ Making nitrogen available to other organisms to absorb

- E.g. **Prochlorococcus** (Most abundant)
 - ⇒ Contributes 13 – 40% oxygen and capture CO₂
- CO₂ Sequestration

ii. **Trichodesmium** : Also a Cyanobacteria

- Do Photosynthesis and Nitrogen fixation concurrently
- Nitrogen Fixation : Only occurs at ↓ O₂
- Heterocyst : Isolated from other cell + Allow to do photosynthesis and nitrogen fixation in one species

(c) Phytoplankton

- Photosynthetic
- Live in **Euphotic Zone**
- Diverse of Shape & Colour : Depending on species and the deepness of habitat
- Living style : Predation + Attack Viruses + Different Nutrient Habit + Different Reproductive Cycles
- Example 1 : Diatoms
 - i. Dominate temperate and polar zones
 - ii. Frustule : Silica Shell
 - iii. Type of Diatoms
 - A. **Centric Diatoms** (Radial Symmetry Top)
 - B. **Pennate Diatoms** (Bilateral Symmetry Bottom)
 - iv. **Have movement** when attach to surface
 - v. Reproduction Steps (Involve both Asexual and Sexual Reproduction)
 - A. Reproduce by cell division (Cut half of themselves)
 - B. Larger half would remain unchanged while Smaller one keep separating
 - C. When the size divided into the limited size, the smallest one will conduct sexual reproduction and form gametes
 - D. Fuse the gametes and the larger part
 - E. Form **Auxospore** : Only synthesize the larger half and give up the smaller one

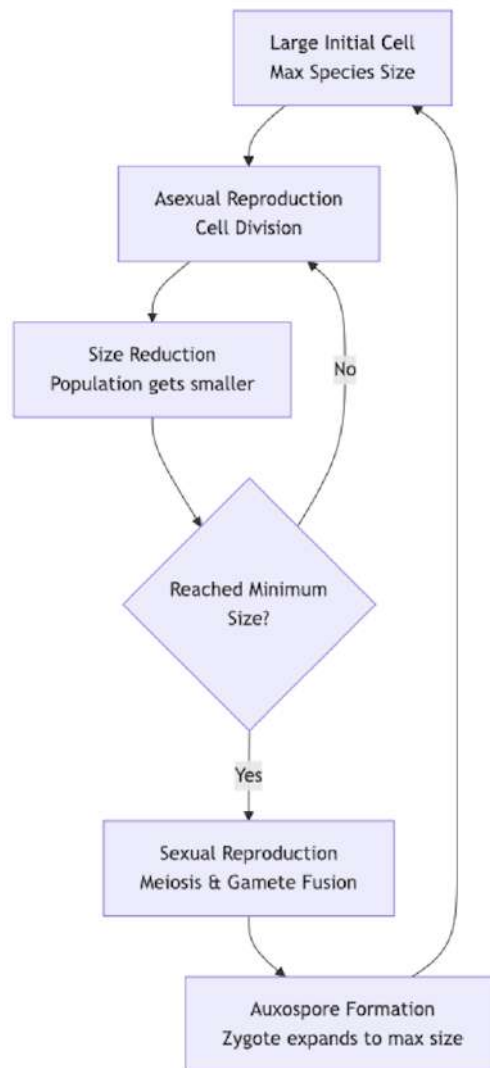


Figure 7: Diatoms Reproduction Cycle

- Example 2 : Dinoflagellates
 - i. Abundant in tropics in summer
 - ii. Two Flagella : Vertical and longitudinal movement
 - iii. **Theca** : Give Distinctive Shape
 - iv. Symbionts for Coral and Invertebrates
 - v. Reproduction
 - A. Asexual by Fission

B. Sexual

⇒ Need to Remove **Theca**

vi. Examples

– Zooxanthellae : Symbiont for corals

vii. Produce **bioluminescence** when mechanically stimulated (Blue Light)

⇒ Usually dominate in one species

⇒ Release Toxins → Affects Nerve Cells → Cause Death

⇒ Bacteria degradation leads to hypoxic condition (Use all oxygen)

⇒ Can affect human's respiration

(d) Mixotrophs

- Consuming bacteria or plankton
- Mainly supplement the low nutrient concentration in certain locations
- Enhance the **energy transfer** between trophic levels

(e) Zooplankton

- Link primary producers and large consumers
- Eat : Phytoplankton, Organic matter, Small Zooplankton
- Have different life stage in their whole life (Classification 2)
- Large migration (Avoid Predation)

(f) Zooplankton : Crustacean Plankton (Copepods)

- Appears in Arctic ocean
- Contain Chitin paired appendage and segmentation
- Very fast swimmer
- Move : Use force of inertia
- Stop : Viscosity
- Can jump out water
 - ⇒ Escape Predation
- Get food : Trap particles with hair-like maxillary
 - ⇒ Have sensory cells in antennae to detect prey and food

(g) Zooplankton : Crustacean Plankton (Krill)

- Larger than **Copepods**
- Dominate the Antarctic ocean and polar food webs
- Live in a large group to avoid predators
- Food : Food particles
⇒ Feed with appendages covered in bristles (setae)

(h) Zooplankton : Gelatinous Plankton

- Can swim but cannot against the water current
- Feed on phytoplankton and small animals
- Example : Jellyfish

i. Zooplankton : Gelatinous Plankton (Jellyfish)

- Meroplankton (Classification 2)
- Nematocysts : Stinging structures in tentacles
- Produce powerful toxic



Figure 8: Jellyfish

ii. Zooplankton : Gelatinous Plankton (Siphonophores)

- Polyp Form
- Live in a group
- Colonial Organism (Stick together)
- Tentacles : Defense + Reproduction

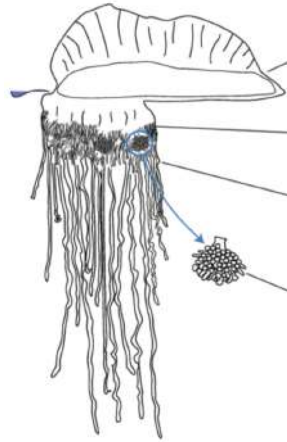


Figure 9: Siphonophores

iii. Zooplankton : Gelatinous Plankton (Ctenophore)

- Transparent (No Colour)
- Long tentacles
- Cilia : Refract light when moving
- Movement : 8 rows of plates with combs

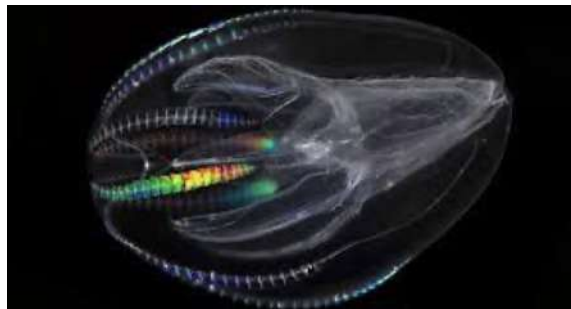


Figure 10: Ctenophore

iv. Other groups

- Larvaceans : Filter feeders
- Salps : Consume particles with mucus

(i) Zooplankton : Ciliate

- Covered by cilia
- Cilia : Help with **feeding and locomotion**

- i. Zooplankton : Ciliate (Foraminifera)
 - Diverse shape
 - Secrete **Calcium Carbonate skeleton**
 - Reproduction : Involve asexual cell division and a cycle of gamete formation
 - Death : Sink to the bottom (Indicator Species)
- ii. Zooplankton : Ciliate (Radiolaria)
 - Tropical water
 - Have Shell (Silica)
 - Have Spine (Avoid Predation and ↑ Flotation)
 - Have mixotrophs symbionts (Photosynthetic Symbionts)
 - Death : Secrete Siliceous Ooze (Form Ocean Sediments + Fossil Record)
- iii. Other Groups
 - Arrow worms : Hermaphroditic + produce toxins
 - Pteropods : Snails live in plankton + Swim by foot + Important in calcium carbonate cycling (sensitive to acidification)

Nekton or free swimming organisms

1. Nekton

- Can swim against ocean current (Can Create Inertia) = **High Reynolds Numbers**
- Powerful swimmers
- Opposite to plankton
- Habitat : Between Plankton and Benthos (Some have Demersal Habit)
- Demersal Habit : Associated to the bottom

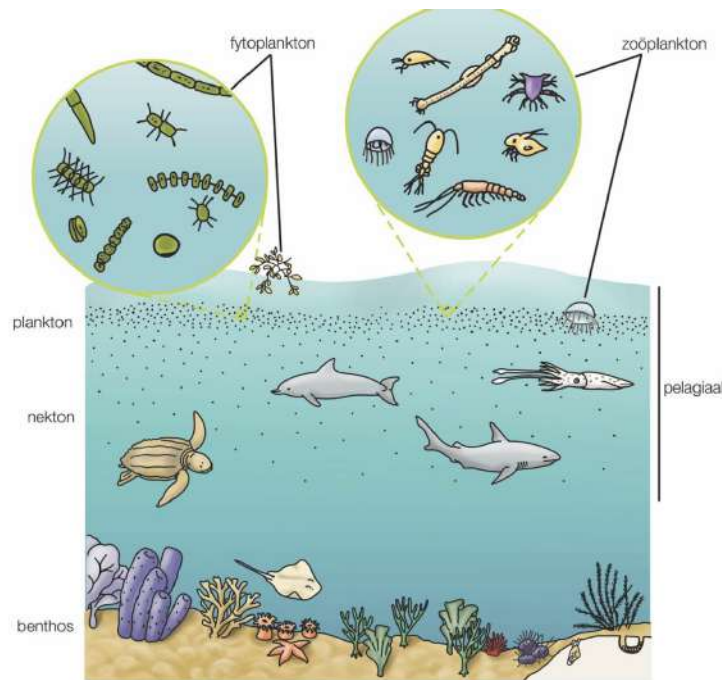


Figure 11: Levels of Habitat

2. Cephalopods (Phylum : Mollusca)

- Carnivorous (Trap prey by Tentacles and Suckers)
- Mouth : Powerful Beak
- Complex Nervous System : Central Brain + Extended Nervous Endings to Arms (Large Neurons)
- Well developed eyes
- Have **Chromatophores** : Pigmented cell are controlled by nervous system → Enlarge and Contracted rapidly
 - ⇒ Function 1 : Attract females
 - ⇒ Function 2 : Adjust colour & texture of skin to background → Prank the Prey (**Camouflage**)
- **Locomotion**
 - (a) Tentacles
 - (b) Jet-Propulsion : Water is pumped into the mantle and moved out through siphon
 - (c) Lateral Fins : Squid

- Buoyancy Control
 - (a) Shell can control the ratio of water and air (Control Flotation) → Nautilus
 - (b) Cuttlebone has small chambers : Lamellae (Squid)
 - (c) Move water by controlling the salt moving through chambers and membranes → Control by osmosis

3. Fish

- Types of fish
 - (a) Jawless Fish : Feed by suction
 - (b) Jawed Fish : Bony Fish + Cartilaginous Fish
- Swimming
 - (a) Body Fusiform Shape : Can Swim Freely
 - (b) Body Odd Shape : Sink to Bottom
 - (c) Fins : Paired + Single Median Fins
 - (d) Fast Swimmers : Tails
 - (e) Slow Swimmers : Elongated body + Require to move their body
 - (f) Sensors
 - Lateral Line System : Detect Movement, Vibration and Water Pressure
 - Found in the side of the body
 - **Neuromasts** : Respond to the water motion
 - **Ampullae of Lorenzini (In Pores)** : Feel Electrical Fields (Found in Sharks)
⇒) Can detect prey hidden in the sand and night
 - (g) Buoyancy : Can save energy
 - Sharks : Fatty liver (Squalene)
 - Bony Fishes : Gas Bladder / Swallow air at surface
 - (h) Feeding
 - Two Sets of Teeth : Holding Prey
 - Large Plates : Grinding Invertebrates (Break the bones)

- Sharp Teeth : Stabbing prey
- **Modified Gill-raker** : Filter Plankton
- Large Fused Teeth : Scraping algae
- Suction Feeding

(i) Reproduction

- Most fish are **Oviparous** = Hatch from an Egg
- External Fertilization
- Rare Internal Fertilization
⇒ E.g. Sharks
- Large Spawning Aggregations
- Ovoviviparity : Egg but come out directly from mother
⇒ E.g. Sharks and Ray
- **Sequential Hermaphroditism** : they have different sex in different phase and cycle (Not 2 sex at the same time)

4. Marine Birds

- Adaptation

- **Monogamous** : One Husband and One Wife
- **Share Caring** for the Young (Incubation + Feeding)
- Breeding Space : **Isolated Beaches** → Avoid Predators
- Powerful Swimmers = Not a Great Divers (Control by the body shape)
- Have **Salt Glands** : Excrete water with high salt
∴ they drink seawater
∴ Intake many salt
- Webbed Feet : Locomotion and Diving

- Examples

(a) Penguins

- Flightless and flippers for swimming
- Habitat : Antarctic and Southern Hemisphere
- Circular System : Countercurrent Heat Exchange
⇒ Heat flow from heat to cold side

- External Parental Care
- (b) Petrel
 - With Hooked Bills (A type of peak)
 - Can detect phytoplankton and fish
- (c) Pelican
 - Only eat fish
 - Good at diving
- (d) Gulls, Terns and Auks
 - Eat small fish and zooplankton
 - Large breeding aggregation
- Feeding
 - Gull : Feeding from Surface
 - Gannet & Pelicans : Plunge Diving
 - Tern : Diving from Air
 - Puffin & Cormorant : Underwater Pursuit

Life in Benthic Habitats

1. Benthic ecosystem

- Plant and animals associated with ocean bottom
- All organic matter sink from water column (Especially Calcium Carbonate)
- Classification 1 : Size : Micro < Meio < Macrobenthos
- Classification 2 : Feeding Habit : Deposit feeder (From sediments) vs Suspension Feeder (Zooplankton) vs Herbivores (Seagrass) vs Carnivores (Macrobenthic prey) vs Scavengers (Remaining plants & animals)
 - ⇒ No Photosynthetic Species (∴ No Light)
 - ⇒ One species can have Multiple Classification

2. Classification 3 : Deepness of Benthic Organisms

- (a) **Epibenthos** : Live on top of sediment
- (b) **Hyperbenthos** : Float above the surface
- (c) **Endobenthos** : Burrowing organisms in sediments

3. Deposit Feeders

- Macrobenthic Organisms
- Get nutrients from sediments
- Habit : Microalgae, Bacteria, Dead Phytoplankton and Dead Sea Grass
- Strategies : Directly Feed, Tentacles, Siphons (Sucking Sand)
- **Surfactant (Detergent)** : Separate Organic Matter from water and break down by **Enzyme**
- Digestion is not efficient (Many are Undigested)
- Some depends on **symbiotic bacteria** in guts and gills
 - ⇒ Obtain Nutrition from sediment
- Anaerobic bacteria can oxidize **Sulfur compounds (Shallow Water) / Methane (Deep Sea)** → Provide organic material to host

4. Suspension Feeders

- Passive Suspension Feeder
 - (a) **Protrude feeding organ** to water current
 - (b) Colonial and Bush-like and form large structure
 - (c) Food Intake : Food is entering from filtering appendages to the gut
- Active Suspension Feeder
 - Create their own current to take food
 - Captured with their mouth, gills, tracts with cilia or mucus layers
 - Can reject unsuitable food particles.
- Food selection : Non Toxic & Not Nutritious
 - ⇒ **Grooves** : Transport particles to gut by filtering the food
- Avoid Clogging (Stuck) : Spit out sand by **Siphon** & Throw away by **Mucus**

5. Benthic Carnivores

- Prey Detection : Vision (Crab), Smell (In the Sand), Sensing Movements (Bobbit Worm)
- Mobile Predators : Can move
- Stationary Predators : Wait prey to pass by
 - ⇒ Adaptation : Large Jaw + Drilling Structure + Venom

6. Benthic Herbivores

- **Microphages** : Feed on microalgae & cyanobacteria by scraping on rocks
 - ⇒ Scrapping structures : Remove material from sediment
- Strategies
 - Radula : Teeth to scrape surface on **rocks**
 - Fused teeth : Scrape microalgae that grows on **coral**
- **Macrophages** : Consume macroalgae or marine plants
 - ⇒ E.g. Sea Urchins, Wood Boring Bivalves (Process cellulose with symbiotic bacteria)

7. Habitat : Mud & Sand Flats

- Lots of chemical interactions
- Huge amount organic matter
- Food webs at surface and inside sediment
- Grain Size : Control by force of the current
 - ⇒ Can control distribution of benthic organism
 - ⇒ Stronger Current = Larger Grain Sizes
 - ⇒ Small one already float away
- Dynamic

8. Burrowing Organism

- Exert great pressure to displace sediment and eliminate water between the grains
- Adaptation
 - (a) Muscular Feed : Able to puncture into the sand
 - (b) Strong Appendage : For digging
- Avoid Predation & Get more Food

9. Oxygen Availability (Key to identify the living layer)

- Upper Layer : Oxygenated (Contact with water)
- Anoxic Layer : Anoxic (Rotten egg smell)→ Achieve anaerobic respiration
 - ⇒ Bacteria : Produce H_2S by using SO_4 in electron acceptor to provide energy
- the boundary between Oxygenated and Anoxic zone is the **Redox Potential Discontinuity (RPD)**
- Methane Layer : Have methanogenic bacteria
- Burrowing organism can bring oxygen to sediments

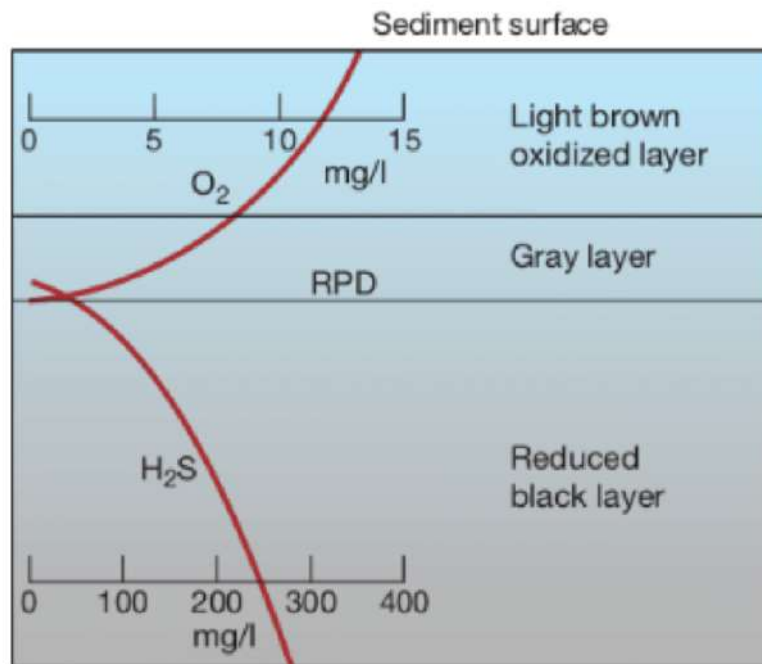


Figure 12: Oxygen in Different Level

5 Lecture 9 & 10 Coastal Intertidal Ecosystems

1. Coastal Intertidal Habitat

- Receive Nutrients : From Terrestrial + High Coastal
- Available food for juveniles
- Fast changing condition
- Affected Factors : Tides & Waves

2. Waves

- Produced by wind (Energy form and move the object from crest to another)
- Tsunamis : Created by underwater volcanic eruptions + Earthquakes
- Wavelength = L = Distance between crests
- Wave Height (Amplitude) = H = Length from Crest to Trough
- Period = T = Time passing through crest to crest
- Velocity = $V = \frac{L}{T}$
- Water Particles move in orbits
 $\Rightarrow \downarrow V \Leftrightarrow \uparrow \text{Depth}$

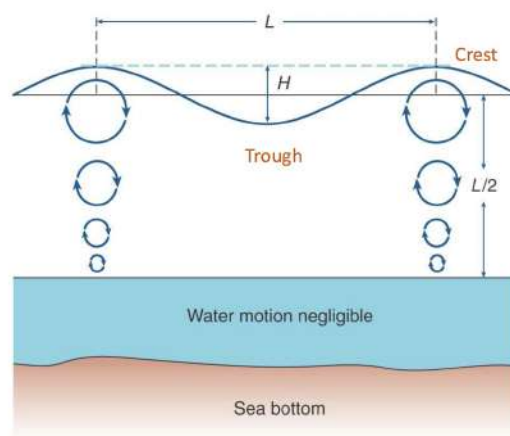


Figure 13: Wave Motion

- (a) Bottom Deepness $> \frac{L}{2}$ = Wave travels freely **without much change**
- (b) Bottom Deepness $< \frac{L}{2}$: Unstable Waves = $\downarrow$ Wave Velocity + $\uparrow$ Wave Height + $\downarrow$ Wavelength
 $\therefore$ Base Wave is slow $\Leftrightarrow$ Top Wave continues moving faster
- (c) Biggest Waves = Quick change of depth and Long wavelengths

3. Longshore Currents : Parallel to Shore

- Crest Waves can **create an angle** from the shore $\rightarrow$ Form a parallel current to shore
- Erosion on shore : Because of **Longshore currents**
- Rip Current : Current that run offshore
 $\Rightarrow \therefore$ Irregular shores

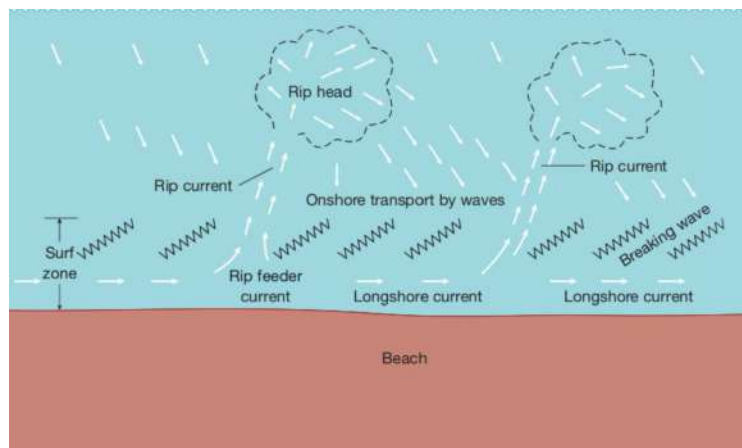


Figure 14: Longshore Current & Rip Current

- Jetties (Pier) also make longshore current
- Wave Refraction
 - (a) Headlands (Rocky point) : Stronger Wave Energy (Convergence) $\rightarrow$ More Erosion

- (b) Sandy Bay : Weaker Wave Energy (Divergence) → Less Erosion = Sand can accumulate = Pattern of accumulation is modified by waves

4. Tides

- Depending on **Gravitational Effects** of sun and moon
⇒ Closer to Earth → Moon has more influence
- It depends on Earth's self rotation
- Moon's gravitation force can pull water ⇔ Opposite side : Earth's Gravity & Rotation produce centrifugal force to the center of planet
⇒ the opposite side also raise and pull up

5. Tidal Variation 1

- **Syzygy** : Earth & Sun & Moon are on the same straight line
- When it is syzygy → Generate **Maximum** Tidal Ranges = **Spring Tides**
⇒ Higher High Tides & Lower Low Tides
- Sun + Earth + Moon form 90° : Gravitational Effects is cancelled each other
⇒ Generate **Minimum** Amplitude = **Neap Tides**
⇒ Tidal Range (Tidal Difference) is small

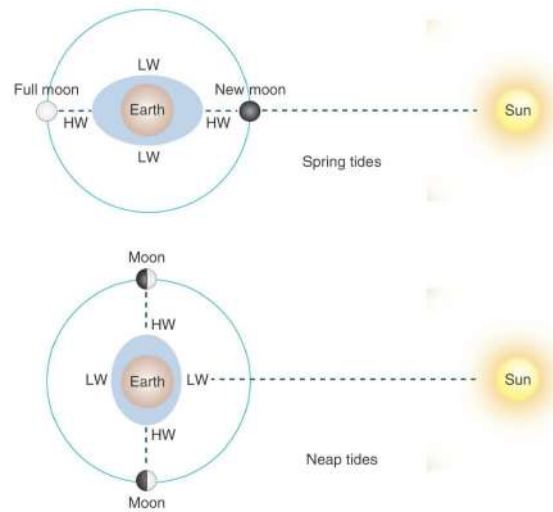


Figure 15: Spring Tides & Neap Tides

6. Tidal Variation 2

- Tidal Day = 24 hours and 50 minutes
- 1 Tidal Day = 2 Low Tides + 2 High Tides
- the actual tides is affected by latitude, basin and Coriolis Effect
- Angle of moon is affected by the equator at different season
⇒ Relative amplitude will change

7. Tidal Variation 3

- Diurnal Tides : 1 High Tide + 1 Low Tide in 1 day
⇒ E.g. Arctic, Indonesia
- Semidiurnal Tides : 2 High Tides (Similar Height) + 2 Low Tides (Similar Height)
⇒ E.g. East US

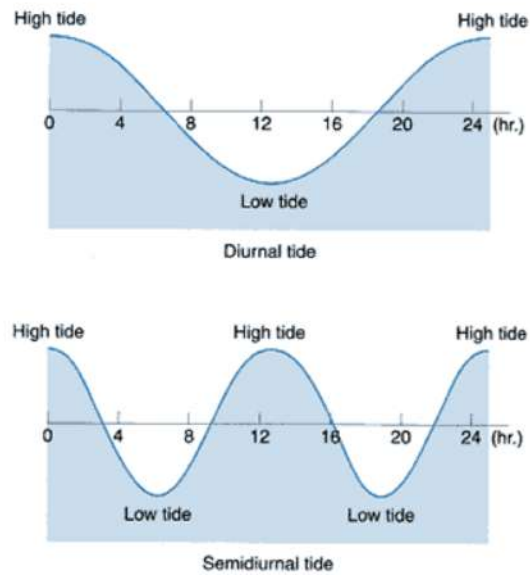


Figure 16: Diurnal Tides vs Semidiurnal Tides

- Mixed Tides : Semi-diurnal and Diurnal combined = 2 High Tides + 2 Low Tides (Not Similar Height)
⇒ E.g. Thailand

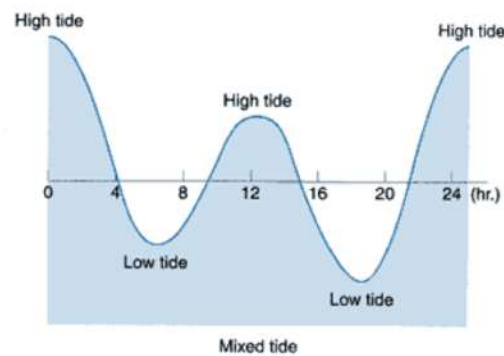


Figure 17: Mixed Tides

- Influence 1 : Time of exposure in air (Affecting the Humidity of mud & rocky shores)
- Influence 2 : Extreme differences of habitat (∴ Dynamic Wave)

8. Rocky Shores & Beaches

- Intertidal Zone : Between Highest and Lowest tide (Vertical Zonation)
- Species Distribution : Horizontal Bands
⇒ Controlled by : Competition, Physiology, Predation
- Similar Environment : Mangroves, Shores

9. Intertidal Zone : Low Tides

- Challenging : Desiccation (Become Dry), Heat, Predation, Absent of food, Lack of Oxygen
- Natural Selection : Reduce metabolic rate, Circulation of body fluid (E.g. Heart Rate), Stay in crevices
- Adaptation : Heat Shock Protein
 - (a) Help with **Protein Folding**
 - (b) ↑ Temperature → Still maintain protein functioning
 - (c) **Only activated** when individual has heat stress
 - (d) Activated for short period (Requires many energy)
⇒ If cannot cool down, it will die
- Species can follow the crashing waves → **Searching Food**
⇒ ∴ Species can sense the change in water pressure and moisture of sediment

10. Interspecific Competition

- ***Chthamalus (CHT)*** vs ***Semibalanus (SEM)***

Factors	CHT	SEM
Species Habitat	Dominant in Upper Intertidal	Mid-tide - Mid-low Neap Tide
Larvae Habitat	Higher (Less Desiccation)	Lower (More Desiccation)
Growth	Weaker	Stronger

Table 1: ***Chthamalus (CHT)*** vs. ***Semibalanus (SEM)***

- Periwinkle : A flower that eat SEM
⇒ SEM cannot reach lower areas of intertidal

- Upper Limit of intertidal : Physiology of species
∴ CHT dominates
- Lower Limit of intertidal : Competition & Predation (SEM vs Periwinkle)

11. Direct & Indirect Effects

- Direct Effects : Predation, Consumption (Negative Interaction)
- Indirect Effects : Helping (Positive Interaction)
⇒ Depending on the specific species
- Example : Ralfsia vs Mussels vs Barnacles vs Morula vs Acanthina vs Limpets Interaction

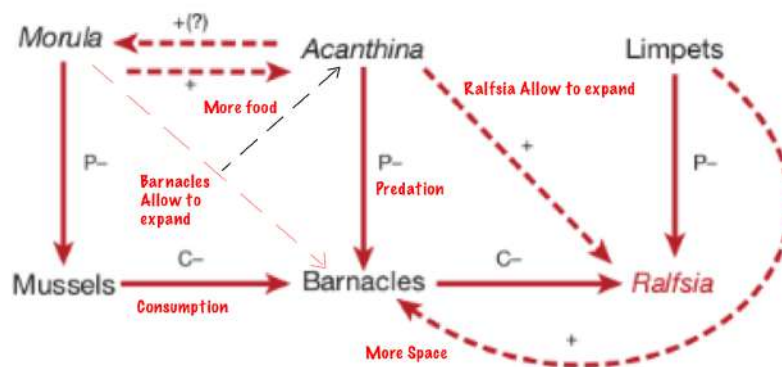


Figure 18: Effects Flow

12. Disturbance and the Effect on Competition

- Mud and Sand : Exposure to rocky shore → Desiccation + Predation
⇒ Animal hide inside the sediments (Moist + Cooler)
- Competition in soft sediments : Space & food & Oxygen
- Vertical stratification in sediment : occupying different depth can minimize the competition
- Can emit toxic compound to keep others away
- ↑ Depth of Sediments ⇔ ↓ O₂ Concentration

- Deposit Feeder have more stable food sources
- Adaptation
 - Collect food deeper in the sediment
 - Collect food at the interface of the sediment and water
 - Eat immediately when you pass through food and sediments

13. Salt Marshes

- Salt tolerant grasses that colonize tidal areas with calm water (No Waves)
- Trap fine sediments + stabilizing shorelines
- **Can Reproduce Sexually**
 - ⇒ But spread very quickly asexually
- Complex Vascularization : Allow exchange of oxygen to roots
 - ∴ sediments are anoxic
- Leaves contain **Silica** to avoid predation
 - ⇒ But flower don't have
 - ⇒ So flower is easily digested
- Vertical Zonation : Each area is dominated by different species
 - ⇒ Controlled by physiological tolerance and competition
- Spartina : An organic matter (Very Invasive) : Convert mud-flats to Vegetated Marsh (Full of Grass)
- Leaves have cellulose and silica
 - ⇒ Hard to be eaten when it is alive
- If die → Bacteria and Fungi can decompose it
 - ⇒ Bacteria become a food source

14. Mangrove Forests

- Large and complex roots on **soft anoxic sediments**
- Limit coastal erosion
 - ⇒ ∴ ↓ Wave Energy + Stabilizing Sediments

- Large predators cannot enter
⇒ Less predation and competition
- Food Web : Leaves is not the directed food source for herbivores
∴ Will fall into the water and degrade by bacteria
∴ Environment is Anoxic
∴ Take long time to degrade
- Bacteria and Fungi are also the food source
- Mangroves = Ecosystem Engineers
 - (a) Provide substratum for multiple benthic species
 - (b) they represent nursery areas (Lay eggs / breeding) for a large number of species
- Adaptation
 - (a) Adaptation 1 : Roots grow upwards & Extending from the trunk
⇒ Allowing Oxygen exchange
 - (b) Adaptation 2 : Salt Tolerance
⇒ Example 1 : Have Salt Gland → Secrete extra salt from leaves
⇒ Example 2 : **Ultrafiltration System of Ion Channel** → Exchange H^+ (Out) + Na^+ (In)
 - (c) Adaptation 3 : **Tannins** in leaves
⇒ Avoid Predation
⇒ Avoid damage from anoxic and iron rich soils
 - (d) Adaptation 4 : Floating Seeds and Propagules
⇒ Seeds germinate when still being attached to parents
⇒ then float to new place and colonize new environment fast
- Keep water is energetic unfavorable
⇒ So require many energy to store freshwater
∴ Fluids in the plant has higher water potential than the surrounding
- Zonation : Opposite to mud & sand

- (a) Upper Limit : Controlled by Competition (Prevent and want to keep away from salt environment)
- (b) Lower Limit : Control by Physiology (Depending on salinity tolerance)
- Mangroves do not use the salt-water
⇒ They just adapt the environments

15. Seagrass Beds

- Shallow Water (< 3m) Marine **Angiosperm** = Flowering Plant
- Large group (∴ **Rhizome System** ≠ Roots)
- Can move (Not stay there for a whole year)
- Distribution : Similar to Coral Triangle Distribution (Very diverse) = Indo-Pacific Coral Triangle
- Function 1 : Stabilize sediments and accumulate particles
- Function 2 : Get N₂ from water and **symbiotic nitrogen-fixing bacteria** in rhizome
- Function 3 : Thin cuticle on the leaves : Allowing gas and nutrient exchange in the water
- Barre Sand = Sand under water bodies
 - (a) Barre Sediments : Lack of nutrients
⇒ Hard to form new colonization
 - (b) Solution 1 : Extend Seagrass by rhizome
 - (c) Solution 2 : Succession (Allow algae to colonize)
⇒ Seagrass invade the environment and wait until the climax community is formed and able to settle
- Food Webs
 - (a) Most organism cannot feed directly from leaves
∴ Lack of symbiotic bacteria to degrade **cellulose**
∴ Need to wait the leave die and degraded by bacteria
 - (b) **Epiphytes (Bryozoans and Microalgae)** grow on the leaves
⇒ Reduce Photosynthesis Capacity

- (c) Invertebrates can scrape the leave to eat the epiphytes
- Seagrass are Threatened
 - (a) Block light (Coastal Development)
⇒ Reducing Photosynthesis
 - (b) Excess nutrient by Human Activity
⇒ ↑ Algae growth & ↑ Phytoplankton → Seagrass has no place to grow
 - (c) Epidemic Diseases = Clones
 - (d) Commercial Shipping

16. Oyster Reef (Filter Feeder)

- Ecosystem engineer in estuaries (River Hole)
- Larvae choose to settle on hard surfaces
⇒ Form large accumulations
- Feed on phytoplankton
- Live in low salinity + Shallow Waters
- Advantages
 - (a) Consume great amount of phytoplankton → Reduce turbidity in coastal areas
 - (b) Influence Nitrogen cycle: Remove Phytoplankton + Remove Nitrogen from water column and Deposit it on benthos → Reduces **Primary Productivity and Levels of O₂**
∴ Avoid algae, phytoplankton Blooms
 - (c) Stabilize Sediments
 - (d) Create a Habitat for Multiple Species

17. Estuaries

- Mixture of salt water & fresh water
- Connected to open sea
- Water Movement : River Discharge + Tidal Action + Basin Shape

- Dynamic (Relative Low Biodiversity), But **calm water**
 - ∴ Low Fetch = Less Wind
 - ⇒ Calm water → Sedimentation
 - ⇒ Sediments → Submerged Aquatic Vegetation (SAV)
- Fetch : Length over water which wind has blown
- High Energy Source → High Productivity
- Sensitive to global change
- Estuary Types (Geomorphology)
 - (a) Coastal Plain
 - (b) Tectonic (via Subduction)
 - (c) Lagoon (Semi-enclosed)
 - (d) Fjord
- Estuary Types (Hydrology)
 - (a) Positive (Salt Wedge) : River Discharge > Tidal Intrusion
 - ⇒ River Dominated (Common)
 - (b) Homogeneous : River Discharge < Tidal Intrusion
 - ⇒ ↑ Evaporation ; ↓ Discharge = Neutral
 - Marine Dominated (Rare)
 - (c) Negative : Evaporation maintains **High Salinity** (Inverse Estuary)
 - (d) Seasonal : Wet / Dry periods cause **Variation**
 - ⇒ Positive → Negative
 - (e) Highly Stratified : Separated Layer due to **Density**
 - ⇒ Often due to salinity or temperature gradients
 - (f) Moderate stratified : Equal Density (Run in **Diagonal Towards the sea**)
 - (g) Vertically Homogeneous Estuary : Vigorous Mixing, **Strong Tidal Control**, Rapid change in Salinity, Mainly in shallow estuaries

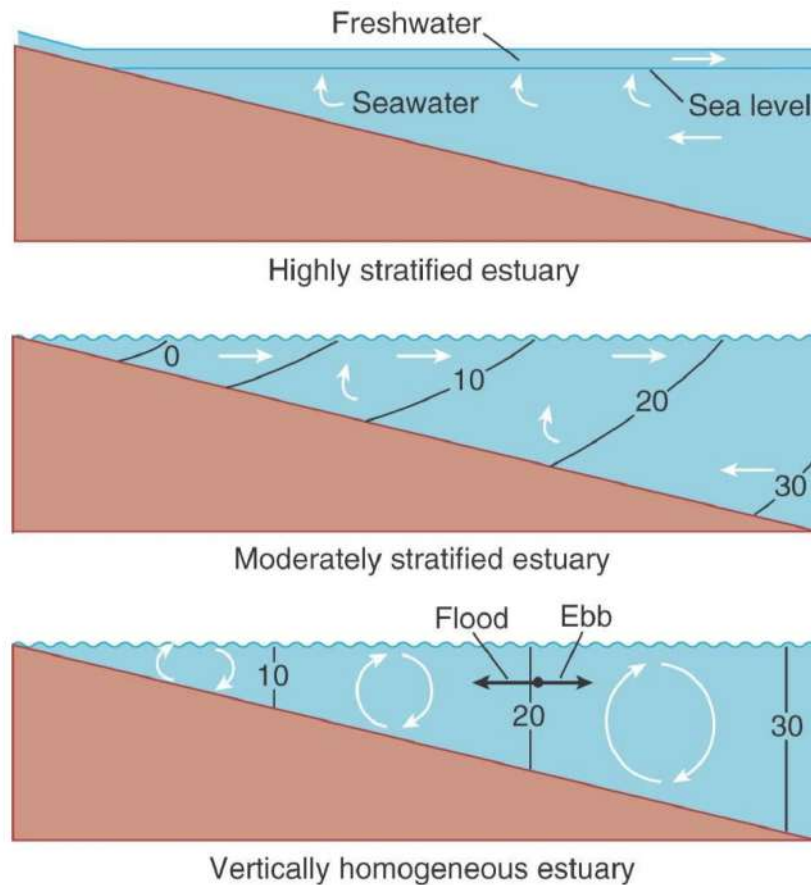


Figure 19: Estuary Type (Hydrology)

18. Substrates

- Large particles sink first (Upper Reaches)
- Fine sediments carried by downstream water current
- Eventually, Arrive the Lower Reaches = Velocity ↓
- **Flocculation** = Salt increase **Particle Binding**
⇒ ↑ Salinity , ↑ Sedimentation , ↓ Turbidity
- Rich Organic & Great Substance → Good for Plants and Deposit Feeders

19. Flocculation

- Fine particles in the suspension **stick together**
⇒ Form a large aggregates

- Larger and Heavier Aggregates can sink first (Deposition)
- Eventually, $\uparrow$ Salinity $\rightarrow$ $\downarrow$ Turbidity

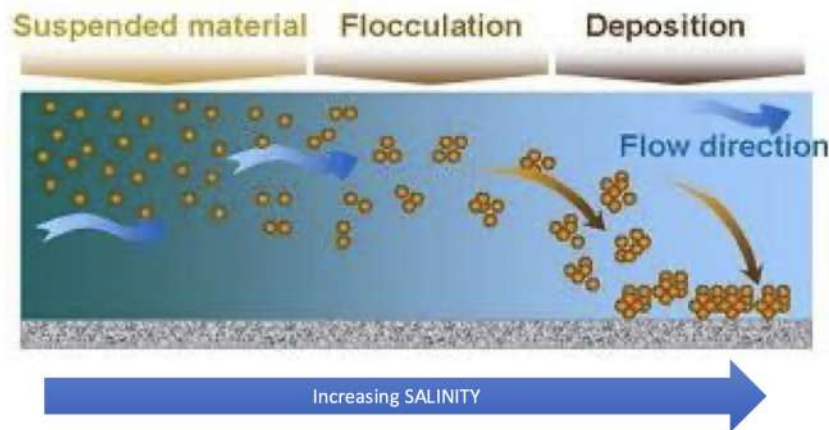


Figure 20: Flocculation

20. Temperature

- Ocean has **High Heat Capacity** : Slow to heat or cool
- High Surface Area relative to Volume Ratio = More Rapid Δ Temperature
- Estuary Freshwater : Colder or Warmer than marine waters
 $\Rightarrow$ Stability : Sea Water $>$ Fresh Water

21. Oxygen

- Recall : $\downarrow$ Temperature, $\uparrow$ Dissolvable in water
- $\uparrow$ Temperature $\Leftrightarrow$ $\downarrow$ Oxygen
 $\therefore$ Gas is less dissolvable in water
- $\uparrow$ Sediment Organics $\Rightarrow$ $\uparrow$ Respiration $\Leftrightarrow$ $\downarrow$ O_2
 $\therefore$ $\uparrow$ Oxygen Demand

22. Salinity vs Species

- **Steno** = Narrow ; **Eury** = Board ; **Haline** = Salt
- **Stenohaline** : most mrine species
- **Euryhaline** : Truly Estuarine

- **Transitional** : Can cross over from salty environment to fresh water environment
⇒ Adaptation : Feed + Reproduce
- **Ana** = Up ; **Cata** = Down ; **Dromus** = Running (Can migrate)
- **Anadromous** : Move from marine to freshwater
⇒ Example : Salmon
- **Catadromous** : Move from freshwater to marine
⇒ Example : Eels

23. Animal Adaptation in Salinity Environment

- Poor Osmoregulators : Protective Shells
- Good Osmoregulator 1 : Impermeable Shell
- Good Osmoregulator 2 : Use **amino acids** to regulate the intracellular osmotic balance
- Good Osmoregulator 3 : Tolerant kidneys

24. Plant Adaptation in Salinity Environment

- Salt Glands : Salt Excretion
- **Aerenchymal Tissue** : Allow gas exchange in **Anoxic Sediments**

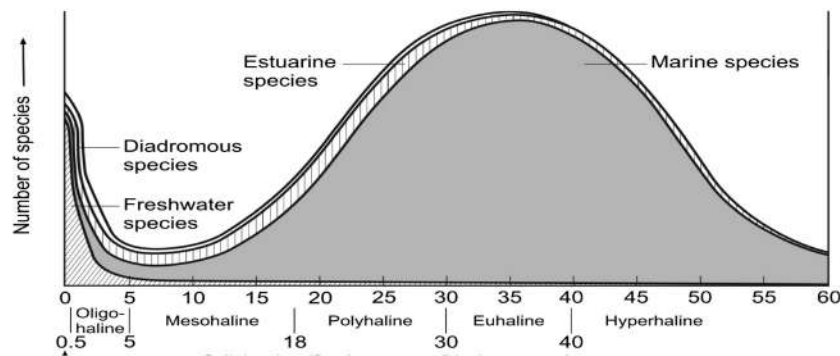


Figure 21: Estuarine Biodiversity

25. Eutrophication : Enriched Nutrient Overload

- ⇒ Excessive growth of simple plant life like algae
- ⇒ Algae might use all the oxygen in water → Species die

26. Eutrophication Overall Flow : Nutrient Overload → Algal Bloom
→ Light Blockage → Plant Death → Bacterial Decomposition
→ Oxygen Depletion → Dead Zones
27. Hong Kong Mai Po Marsh : A Wetland
⇒ Allow species to rest, reproduce and migrate

6 Lecture 11 Marine Biodiversity

1. Biological Diversity

- The variability among living organisms from different ecosystems and ecological complexes
- Include diversity within species, between species and of ecosystem
- No static
- Genetic Diversity, Species Diversity, Functional and Tropical Diversity, Ecosystem Diversity

2. Genetic Diversity : The Building Blocks

- The variation of genetic material within and among individuals of a species
- **Allelic Variation** within population
- Difference in DNA sequences
- Unique genotypes **adapted** to local conditions
- Determines how species respond to environmental stressors
⇒ Climate Change, Pollution, Ocean Acidification, Disease Outbreak
- Essential Reason 1 : Population Resilience
⇒ E.g. Coral Adapting to thermal stress
- Essential Reason 2 : Selective Breeding in aquaculture
- Essential Reason 3 : Conservation of endangered species at risk of inbreeding depression

3. Species Diversity : The visible Web of Life

- Species Richness : Number of different species
- Species Evenness : Relative Abundance
- Taxonomic and phylogenetic diversity
- Ecological Importance : Loss of Species → Disrupt food webs + Weaken Ecological Interaction + Reduce the system's ability to respond to disturbance

4. Functional and Trophic Diversity

- Food Web : Long, Intricate, Sensitive to Species Loss
- Functional Redundancy : Multiple species performing similar roles → Adds Resilience
⇒ Loss of keystone and foundation species → Ecosystem Collapse

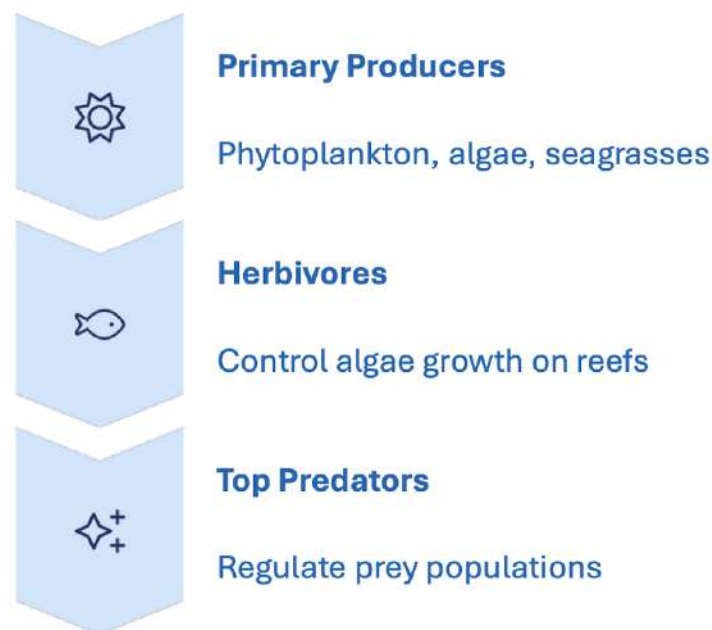


Figure 22: Food Web

5. Ecosystem Diversity : Habitats and Communities

- The variety of ecosystem in a region
⇒ Biological Communities, abiotic factors (Salinity, Depth, Temperature) that shape them
- Examples
 - (a) Coral Reef
 - High Productivity and species richness
 - Vulnerable to bleaching and climate change impact
 - (b) Mangrove Forests
 - Crucial for coastal protection

- Crucial for juvenile fish species
- (c) Deep-sea Vents
 - Support Extremophiles
 - Chemoautotrophic life in harsh condition
- Fundamental to Human Well-beings

6. Threats to Marine Biodiversity

- (a) Overfishing : Depletes fish stocks, Disrupts marine food webs, Decline of marine species
- (b) Pollution : Plastic Waste, Chemical Runoff, Contaminates Marine Habitats, Harms Marine Life
- (c) Climate Change : Cause ocean acidification, Raise sea temperatures
⇒ Negatively impact coral reefs and marine ecosystems

7. Plastic Pollution

- Mistaking it for food
⇒ Leads to injury, starvation, death
- Entangled in plastic waste
⇒ Fishing Nets, Six-pack Ring (Coke)
⇒ Cause physical harm / Drowning
- Damage Critical Habitat : Coral Reefs damaged → Affecting numerous marine organisms
- Accumulation of plastic : Reduce biodiversity + Undermining the health and resilience of marine environment

8. Climate Change on Coral Reefs

- Coral bleaching : Expel symbiotic algae due to stress
∴ Sea Temperature + Ocean Acidification
- Algae provide nutrition + Colour
- Bleaching Corals : Easy to get disease and decline

9. Importance of Marine Biodiversity

- (a) Ecological balance

- Maintain food web and nutrition cycles
- Diverse species interaction : Support ecosystem function
- (b) Economic Value
 - Fisheries, Tourism, Biotechnology
 - Provide resources for economic activities
- (c) Human Well-being
 - Marine biodiversity support human health by food, medicine, and recreational opportunities
 - Climate Regulation and disaster risk reduction
- (d) Sustainable Fishing (Bering Sea Pollock Fishery)
 - Strick Quotas
 - Bycatch Reduction Technologies
 - Ecological Monitoring
 - ⇒ Ensure pollock populations remain healthy and protected
- (e) Sustainable Fishing (North American Lobster Fishery)
 - Regulate Trap Limits
 - Seasonal Closures
 - Size Limits
 - Bycatch Reduction to protect juvenile lobsters
- (f) Tourism and Blue Economy : Combine biodiversity + Tourism

10. Natural Coastal Defenses

- Provide vital ecosystem services by acting as natural users against extreme weather
 - ⇒ E.g. Mangrove, Coral reefs, Seagrass beds
- Stabilize shorelines : Trapping sediments, Reducing Erosion and storm surge impacts
- Coral Reef break wave energy offshore
 - ⇒ Reduce the storm force before reaching the coast

11. Protecting Strategies

- Make a marine protected areas (MPAs)
⇒ Shelter critical habitats + Allow ecosystems to recover
- Sustainable Fishing : Prevent overfishing + Ensure long term fish population health
- Implementing Pollution : Control measures to reduce plastic waste, chemical runoff, and contaminants
- Restoring damaged ecosystems : Enhance biodiversity and resilience

12. Largest MPAs : French Polynesia's marine protected area
⇒ Fishing and Mining are prohibited

13. Temporal and Spatial Dimension

(a) Geological Timescales : Generated by speciation and lost through extinction over a long period of time

(b) Ecological Timescales : Climate Change + Invasion

(c) Spatial Distribution

- Hotspots Exist in tropical reefs : Higher diversity
- Polar and deep-sea ecosystems : Unique, endemic, ancient lineages

(d) Diversity Metrics : Species composition changes across habitats or regions

14. **Alpha Diversity** : How many different type of organisms in one sample

⇒ the higher α value, the more species

- Species Richness : Just **the number of different species present**
- Shannon Index : Accounts for **both richness and evenness**
⇒ **How balanced the species are**
- Simpson Index : Focuses more on dominance (**the most abundance species**)
⇒ Whether a few species are very common
⇒ **High Probability = Dominant Species occurs ⇔ Low Diversity**

⇒ Low Probability = Multiple Species occurs ⇔ No Dominant Species ⇔ High Diversity

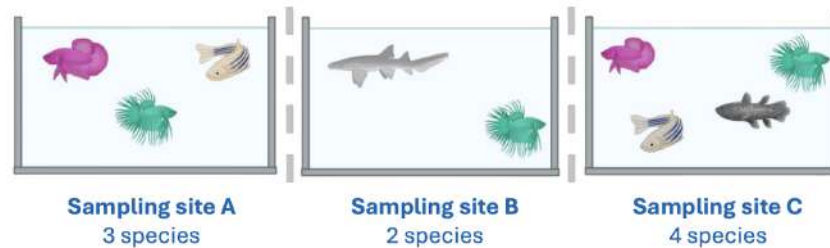


Figure 23: Alpha Diversity

15. **Beta Diversity** : the difference in species composition between two or more places

(a) Dissimilarity Indices : Compare species presence and absence or abundance between samples

- Jaccard Index : Based on **presence and absence**
⇒ Use when you care **if a species is there, not how much of it**
- Bray-Curtis Dissimilarity : Based on **abundance**
⇒ Use when you care about **how many individuals of each species are present**

(b) **Ordination Method (For Visualization)** : Help to see clusters or **patterns of similarity and difference between communities**

- PCoA : Principal Coordinates Analysis
- NMDS : Non-metric Multidimensional Scaling

16. **Gamma Diversity** : Measure the **overall diversity** within a broad spatial scale

⇒ E.g. Ecosystem, Region, Country

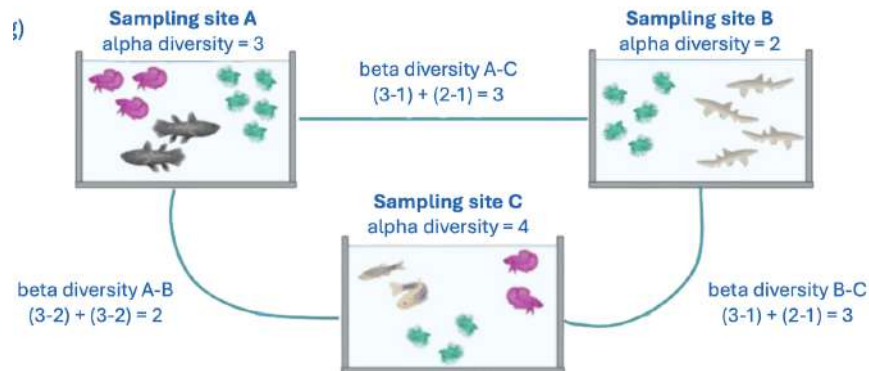


Figure 24: Alpha & Beta & Gamma Diversity

17. Methodology : Autonomous Reef Monitoring Structure (ARMS)

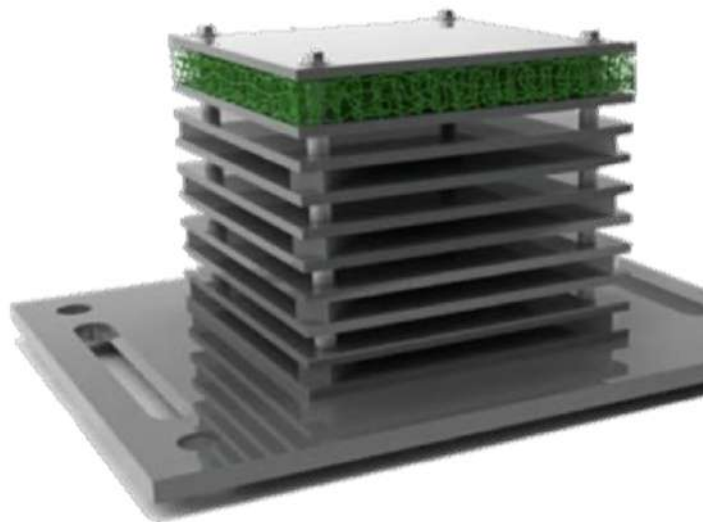


Figure 25: ARMS